

MSML 641

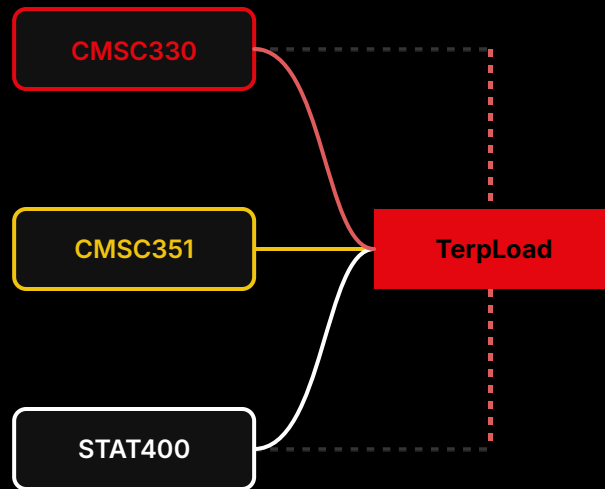
TERPLOAD

Know your semester's workload before you commit

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<https://github.com/moatazsaad/msml641-project>

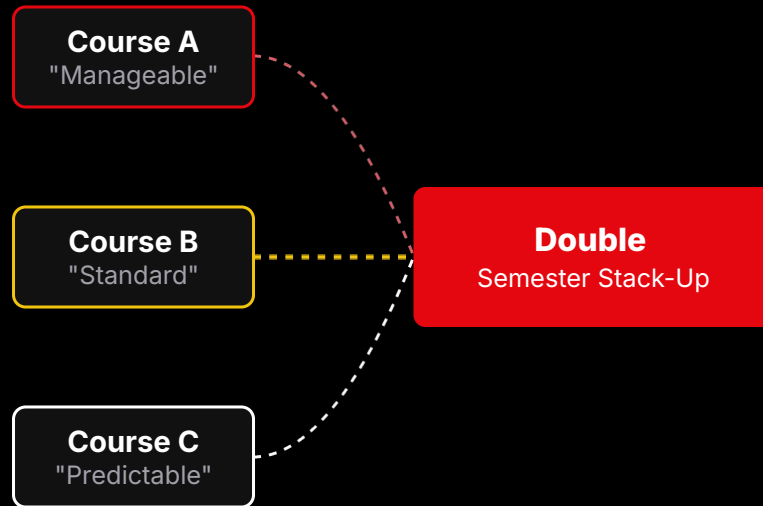
PROBLEM & USERS

The Overlapping Workload Stack-Up

Target user: UMD CS and STEM students planning multi-course semesters.

A course may look manageable by itself. The real problem is when projects, exams, homework, and time demands overlap across several courses.

Students manually compare PlanetTerp reviews, Reddit posts, GroupMe messages, and advice from friends.



EXAMPLES OF STUDENTS' QUESTIONS

Feedback on schedule?

Academic



Y'all got any feedback on how my schedule is looking like as an incoming freshman majoring in CS and in College Park Scholars? I'm a little worried about how little time I have in between MATH140, FMSC190S, and CMSC131 to walk to class. Is it doable?

Is this schedule too ambitious for a freshman?

Academic



gen eds mostly done with AP creds

From Course Selection to Schedule Decision

How TerpLoad works

2. Fetch Reviews

PlanetTerp course reviews are retrieved

4. Course Signals

Review predictions become course-level signals

project · exam
homework · time

6. TerpLoad Report

Risk, confidence, evidence & best move

02

04

06

01

03

05

1. Select Courses

Student enters 1-6 UMD courses

3. DistilBERT NLP Inference

Multi-label review classification

5. Schedule Risk

Signals combine into overall schedule risk

Recent Professor Context

Historical Grades

Demo

Video

NLP Method Evolution

STAGE 1 — RULES / LEXICAL MATCHING

Keyword Baseline

"There were **projects** almost every week and the **final** was **difficult**."

- Simple keyword/rule matching
- Could detect explicit workload language
- Failed on context, negation, and ambiguity



STAGE 2 — STATISTICAL NLP BASELINE

TF-IDF + Logistic Regression

Review → **TF-IDF Features** → LogReg → Probabilities

- Converts **words/n-grams** to weighted features
- Learns from labeled reviews, not manual rules
- No deep understanding of context/order



STAGE 3 — CONTEXTUAL TRANSFORMER

Fine-Tuned DistilBERT (Final)

Raw → **TOKENIZER** → **SUBWORD TOKENS** (256) → **TRANSFORMER** → **MULTI-LABEL HEAD**

- PROJECT-HEAVY

EXAM-HEAVY

HOMEWORK-HEAVY

TIME-CONSUMING
- Captures context beyond isolated words/n-grams
 - One review can receive multiple workload labels
 - Fine-tuned on our labeled PlanetTerp corpus

DistilBERT Won on Held-Out Data

Task: Multi-label classification of student reviews into four workload signals

5,059

PlanetTerp Corpus

Total reviews utilized for model development

70 / 15 / 15

Data Split

Train / validation / test division

493

Labels Corrected

Manual audit of noisy weak labels

MODEL

MACRO F1

MICRO F1

TF-IDF + Logistic Regression

0.652

0.667

DistilBERT (Fine-Tuned)

0.739

0.738

▲ +0.087 Macro F1 (+13.3% relative) vs. TF-IDF

■ **Held-out test set:** 760 reviews held out from training and threshold selection.

WHERE IT FAILS

- **Qualified language:** reassuring phrases can still be over-flagged.
- **Sparse-review amplification:** 1 positive / 3 reviews = 33% (crosses 30% threshold).
- **Mixed reviews:** one review may discuss exams, projects, homework, and time together.

USER EVIDENCE

User Feedback Changed the Product

14 students evaluated our prototype workload report before the final app was built.

14 / 14 Users

found the report format clear



Helpful for Semester Planning

Participants confirmed the prototype report would aid their academic planning.

9 / 14 Users

wanted professor context



Added Recent Professor Context

Directly integrated context links to PlanetTerp to support this specific workflow.

3 / 14 Users

asked about missing evidence



Added Fallback Status Tags

Shipped explicit fallback tags: **NO COURSE REVIEWS**, **LIMITED DATA**, and **UNCERTAIN**.

User Group

wanted deeper decision metrics



Added Historical Grade Context

Injected objective grade distributions while strictly separating historical grade distributions from workload risk.

Prototype feedback directly shaped the final product.

Ethics & Limitations

DATA RIGHTS & PRIVACY

- Public PlanetTerp review data
- No student login required
- TerpLoad does not collect personal student data
- API caching reduces unnecessary requests

BIAS & REPRESENTATION

- PlanetTerp reviewers are self-selected
- Review volume differs substantially by course
- Weak-label creation can introduce labeling bias
- Student experiences may differ by professor or semester

WHEN THE MODEL IS WRONG

- Low-review courses receive explicit warnings
- No-review courses do not receive workload predictions
- Supporting review excerpts let users inspect the evidence
- Historical grades are context only and never affect schedule risk